Demo Blaze Product Store - Manual Testing Project Report

Prepared by: Shubham Ravikumar Narwade | Timeline: 09/2026 – 10/2026

1. Project Overview

The Demo Blaze Product Store is an e-commerce web platform designed to facilitate product browsing, category filtering, user authentication, and online purchasing. This manual testing project was undertaken to validate end-to-end functionality, ensure compliance with baseline e-commerce business logic, and confirm a seamless user experience across the primary workflows.

Objectives

- Verify that user-facing modules operate in alignment with defined functional requirements.
- Identify discrepancies, visual bugs, broken links, and functional failures prior to system deployment.
- Ensure UI stability, cross-browser compatibility, and robust session handling for authentication and cart persistence.

2. Testing Scope & Methodologies

The quality assurance phase incorporated multiple testing tiers to evaluate the platform from smoke verification through to detailed functional and non-functional assessment.

Methodologies Applied

- **Smoke Testing:** Executed at initial build acceptance to ensure foundational features—such as site availability, navigation routing, and modal dialog responses—are operational.
- **Sanity Testing:** Conducted following minor updates and bug fixes to quickly verify that key transactional paths remain stable without regressions.
- **Functional Testing:** Thorough verification of input controls, forms, dynamic category rendering, and state management against business specifications.
- **Black-Box Testing:** Test scenarios designed and executed without internal code visibility, assessing functionality purely from an end-user perspective.

- **Regression Testing:** Systematic re-testing of unchanged application modules following bug fixes to prevent inadvertent regressions in existing functional flows.

Scope Matrix

Module / Scope Area	In Scope	Out of Scope	Description
Authentication & Profile	Yes	No	Sign-up, login validation, session state, log-out.
Catalog & Navigation	Yes	No	Carousel banner, category selection (Phones, Laptops, Monitors).
Cart & Checkout	Yes	No	Add to cart, cart item deletion, purchase modal, receipt generation.
Payment Gateway Backend	No	Yes	Actual banking clearing and live fund transaction processing.
Performance Under Load	No	Yes	High-volume stress testing and multi-user concurrency limits.

3. Test Execution

Test scenarios and granular test cases were established across all primary user journeys, ensuring full coverage from landing to post-purchase confirmation.

Execution Summary by Journey

- **Home & Navigation:** Validated responsive layout, top navigation menu routing (Home, Contact, About Us, Cart, Log in, Sign up), category filters, and product card rendering.
- **Account Management & Authentication:** Tested validation criteria for Sign-up and Log-in modals, including blank inputs, duplicate usernames, invalid credentials, and session persistence.

- **Product Discovery:** Verified dynamic product list population across Phones, Laptops, and Monitors, as well as product details views (title, price, description, and image display).
- **Cart Operations:** Validated addition of products to cart, calculation of total order amounts, persistent table state, and deletion of individual cart entries.
- **Order Placement & Purchase:** Executed end-to-end checkout flow via the purchase modal dialog, confirming mandatory form validation (Name, Country, City, Credit Card, Month, Year) and final order summary modal presentation.

Test Execution Metrics

Journey / Module	Total Test Cases	Passed	Failed	Blocked	Execution Status
Home & Navigation	14	14	0	0	Completed
Sign Up & Login	18	16	2	0	Completed
Product Details & Categories	16	15	1	0	Completed
Cart Management	12	11	1	0	Completed
Order & Checkout Flow	15	13	2	0	Completed
Contact & Support Forms	8	8	0	0	Completed
Total	**83**	**77**	**6**	**0**	**92.8% Pass Rate**

4. Defect Management

Identified defects and recorded them by severity.
Verified actual results against expected results.
Tracked defect status through retesting and resolution.

Defect Log Summary

Defect ID	Module	Scenario / Description	Expected Result	Actual Result	Severity	Status
BUG-001	Account	Register with existing username	Display error message "This user already exist."	Error message displays intermittently; modal remains open	Medium	Fixed / Retested

Defect ID	Module	Scenario / Description	Expected Result	Actual Result	Severity	Status
BUG-002	Cart	Delete item from cart	Total price updates immediately upon row removal	Total price remains unchanged until manual page refresh	High	Open
BUG-003	Checkout	Submit empty purchase form	Form fields trigger validation highlighting required inputs	Purchase completes with blank details and displays empty confirmation	High	In Review
BUG-004	Product	Navigate back from item view	User is returned to previous scroll position on homepage	View resets to category top and resets active filter	Low	Closed

Defect Severity Distribution

- **Critical / Blocker (0):** No severe failures preventing full application usage or system collapse.
- **High (2):** Issues affecting cart totals calculation dynamically and missing client-side purchase form validation.
- **Medium (3):** Intermittent alert rendering and minor navigation anomalies.
- **Low (1):** UI pagination and persistent filter retention post page return.

5. Conclusion & Recommendations

The manual testing lifecycle conducted between September 2026 and October 2026 confirmed that the Demo Blaze Product Store exhibits solid baseline functionality across core consumer workflows. The application demonstrates stable UI responsiveness, reliable catalog rendering, and functional purchase confirmation paths.

Prior to production release, it is recommended to:

- Address high-priority validation bugs within the checkout submission modal to ensure required billing details are mandatory.
- Implement real-time DOM balance updates when removing cart items to avoid calculation discrepancies.
- Expand regression suites into automated test scripts to ensure sustainable quality in subsequent release iterations.